Derivatives: Contracts, Models, and Markets · Technical Note 19

Where the Implied Density Comes From

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Section 19.2 of Derivatives: Contracts, Models, and Markets obtains the risk-neutral density from option prices in a single paragraph: a digital is the limit of a call spread, an infinitesimal butterfly is the limit of a butterfly spread, and its price is therefore the state price for one outcome. That is the argument, and it is the right length for the main text. This note supplies what the paragraph compresses: the derivation in two differentiations, the discrete form that Example 19.2 actually computes, the sense in which nonnegativity of the density is *the convexity condition of Section 11.3, and the three places where the construction meets a real option chain and stops being exact. The last of these is the one that decides published numbers.*

1 TWO DIFFERENTIATIONS

Let g be the density of S_T under the risk-neutral measure and let $c(K)$ be the price today of a European call struck at K and expiring at T. Risk-neutral valuation says the price is the discounted expected payoff,

$$c(K) = e^{-rT} \int_K^\infty (S - K)\, g(S)\, dS \tag{TN19.1}$$

and everything follows from differentiating this twice in the *strike*. The strike is not a state variable, so no stochastic calculus is involved; these are ordinary derivatives of an integral whose lower limit happens to be the variable of differentiation.

The first derivative is the digital. Differentiating (TN19.1) under the integral sign, the boundary term vanishes because the integrand $(S - K)$ is zero at $S = K$, leaving

$$\frac{\partial c}{\partial K} = -e^{-rT} \int_K^\infty g(S)\, dS = -e^{-rT}\big[1 - G(K)\big] \tag{TN19.2}$$

where G is the cumulative distribution function. This is Section 12.8's result: a digital paying \$1 above K is worth $-\partial c/\partial K$, and that price is the discounted risk-neutral probability of finishing above K. Nothing in the main text needs more than this.

The second derivative is the density. Differentiating (TN19.2) once more,

$$\frac{\partial^2 c}{\partial K^2} = e^{-rT} g(K) \qquad \Longleftrightarrow \qquad g(K) = e^{rT}\, \frac{\partial^2 c}{\partial K^2} \tag{TN19.3}$$

which is the Breeden–Litzenberger result quoted as equation (19.1). ■

Three remarks follow, each at a place where a reader imports something the derivation does not contain.

No model has appeared. Equation (TN19.3) is a consequence of risk-neutral valuation alone. It does not assume lognormality, constant volatility, continuous paths, or any particular dynamics, which is exactly why it can be applied to Black–Scholes

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prices to show that the density they imply is not the one the Black–Scholes–Merton model assumes.

The density is of S_T under Q, not under the real-world measure. It is the market's pricing distribution, and it differs from the forecast distribution by the risk premium of Section 19.10. Anyone who takes g as a forecast will overstate the market's assessment of crash probability, because part of the left tail is the price of insurance rather than the chance of loss.

A complete chain is required, in principle. Equation (TN19.3) needs $c(K)$ for every K in $(0, \infty)$. Section 4 is about what happens when it is available at fifteen strikes.

2 THE DISCRETE FORM

Example 19.2 of Derivatives: Contracts, Models, and Markets does not differentiate anything. It takes second differences of quoted prices, and the connection is worth making explicit because it is the form every practical calculation uses.

For strikes evenly spaced at h, the central second difference approximates the second derivative:

$$\left.\frac{\partial^2 c}{\partial K^2}\right|_{K_i} \approx \frac{c_{i-1} - 2c_i + c_{i+1}}{h^2} \qquad \Longrightarrow \qquad g(K_i) \approx e^{rT}\,\frac{c_{i-1} - 2c_i + c_{i+1}}{h^2} \tag{TN19.4}$$

The numerator is not an abstraction. *It is the price of a butterfly*: long one call at K_{i-1}, short two at K_i, long one at K_{i+1}. That structure pays h at K_i, zero at and beyond both wings, and interpolates linearly in between, so its payoff is a triangle of base $2h$ and height h, with area h^2. Dividing by h^2 normalizes the triangle to unit mass, and compounding forward converts a price into a probability density. Example 19.2's fourth and fifth columns are exactly these two operations.

The approximation error is $O(h^2)$, by the usual Taylor argument. For strikes \$2.50 apart on a \$48 stock this is small against the errors of Section 4, which is the reason the main text does not dwell on it.

3 NONNEGATIVITY IS THE CONVEXITY CONDITION

A density cannot be negative. By (TN19.3) that is the statement

$$\frac{\partial^2 c}{\partial K^2} \geq 0 \tag{TN19.5}$$

which says the call price is a convex function of the strike, and convexity in the strike is precisely the butterfly condition of Section 11.3. The two are the same requirement seen from two directions. Section 11.3 derives it from a no-arbitrage argument about a portfolio that cannot have negative value; equation (TN19.3) derives it from the fact that probabilities are nonnegative. Neither is prior to the other.

This is why Section 19.5 can turn a statement about densities into a constraint a quoted surface must satisfy, and why a violation is detectable without any model: a negative butterfly price *is* a negative probability, and one can be read off a screen.

The companion conditions have the same character. Monotonicity in the strike, $\partial c/\partial K \leq 0$, is (TN19.2) restated with $G \leq 1$; a call spread cannot cost less than nothing.

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Nondecreasing total variance across maturities is the calendar-spread condition, and it is not a statement about g at a single T but about the consistency of the family, which is why Section 19.5 treats it separately.

4 WHERE IT MEETS A REAL CHAIN

Three things separate (TN19.3) from a number one can publish, and they are ordered here by how much damage they do.

Strikes are finite and discrete. The integral in (TN19.1) runs over all positive strikes; a chain quotes perhaps fifteen. Everything outside the quoted range must be supplied by an assumption, and the assumption is not innocuous: the $1/K^2$ weighting of the variance strip in Section 19.3 gives low strikes more weight than high ones, so the left wing, the part quoted worst, carries the most influence on a variance-swap or volatility-index value. *Any calculation that integrates a smile beyond its quoted range must state how the smile was extended*, because the extension, not the data, can determine the result.

Prices are quoted with a spread. The second difference in (TN19.4) is a difference of differences, so it amplifies quoting noise. With a bid–ask half-width of ϵ on each of three strikes, the butterfly price carries an error of up to 4ϵ against a signal of order gh^2e^{-rT}. For strikes far from the money, where g is small and spreads are wide, the ratio can exceed one, which is how a chain that is arbitrage-free in mid prices can show a negative butterfly in bids, and why practitioners smooth in implied volatility rather than in price.

The density is a derivative, and differentiation is unstable. Two smooth interpolations of the same fifteen points can agree everywhere in price and disagree materially in second derivative. This is the generic ill-posedness of differentiating noisy data, and it is the reason the surface-fitting apparatus of Section 19.5 exists at all: one does not fit a density, one fits a smile and differentiates something that was built to be differentiated.

None of this weakens the result. Equation (TN19.3) is exact, and the market's distribution really is in the prices. What the three items say is that recovering it is an estimation problem with a stated procedure, and that two analysts who publish different densities for the same chain are disagreeing about the procedure, not about the theory.

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